

Test 6 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. Select **all** the expressions with a value of 25.

- ☐ 5×5
- ☐ 2^5
- ☐ 5^3
- ☐ 25×1
- ☐ 5^2
- ☐ 5×2

Select all that apply.

Show your work

calculator · multiselect

2. The line plot shows books read last month.

Select **all** the values that were reported by exactly 2 students.



- ☐ 5
- ☐ 4
- ☐ 3
- ☐ 7
- ☐ 2
- ☐ 1
- ☐ 8
- ☐ 0
- ☐ 6

Select all that apply.

Show your work

multiselect

3. Select **all** the expressions with a value of 45.

- ☐ $(-5) \times (-9)$
- ☐ $(-9) + (-5)$
- ☐ $(-9) \div (-5)$
- ☐ $(9) \times (-5)$
- ☐ $(-9) \times (-5)$
- ☐ $(9) \times (5)$

Select all that apply.

Show your work

calculator · multiselect

4. A 6-pound box costs \$18.00.

Select **all** the expressions that give the cost per pound, in dollars.

- ☐ $18 - 6$
- ☐ 18×6
- ☐ $6 \div 18$
- ☐ $18/6$
- ☐ $36 \div 12$
- ☐ $18 \div 6$

Select all that apply.

Show your work

multiselect

5. A data set is shown.

0, 1, 1, 2, 3, 4, 5, 6, 9, 10, 11, 14, 14, 14, 26, 26, 26, 27

Select **all** the histogram intervals that would contain exactly 0 values.

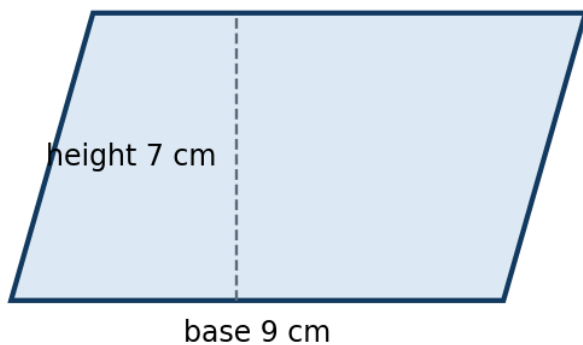
- ☐ 15-19
- ☐ 0-4
- ☐ 5-9
- ☐ 10-14
- ☐ 20-24
- ☐ 25-29

Select all that apply.

Show your work

multiselect

6. What is the area of the parallelogram shown, in square centimeters?

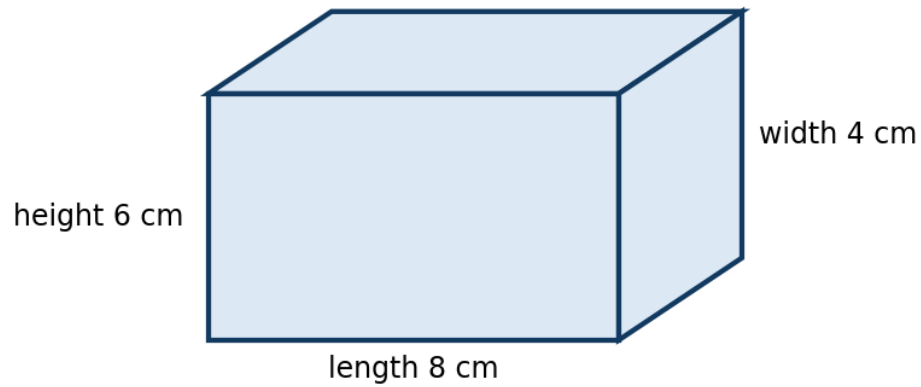


- A. $31 \frac{1}{2}$
- B. 32
- C. 63
- D. 16

Show your work

multiple choice

7. What is the volume of the right rectangular prism shown, in cubic centimeters?



- A. 32
- B. 192
- C. 18
- D. 208

Show your work

multiple choice

8. Select **all** the ratios equivalent to 28:36.

- ☐ 9:7
- ☐ 7:11
- ☐ 35:45
- ☐ 8:9
- ☐ 7:9
- ☐ 42:54

Select all that apply.

Show your work

multiselect

9. A data set is shown.

14, 14, 14, 17, 18, 19

The value 29 is added to the data set.

Select **all** the measures that change.

- ☐ Maximum
- ☐ Median
- ☐ Mean
- ☐ Range
- ☐ Minimum

Select all that apply.

Show your work

multiselect

10. What is $0.6 \div 0.05$?

- A. 0.03
- B. 1.2
- C. 120
- D. 12

Show your work

calculator · multiple choice

11. A data set is shown.

8, 11, 13, 15, 16, 21

Select **all** the statements that are true about the data set.

- ☐ The range is 17.
- ☐ The median is 16.
- ☐ The mean is 14.
- ☐ The median is 14.
- ☐ The range is 13.
- ☐ The mean is 17.

Select all that apply.

Show your work

multiselect

12. Select **all** the expressions with a value of 6.

- ☐ $21 - (15)$
- ☐ $21 + (-15)$
- ☐ $21 - (-15)$
- ☐ $21 \times (-15)$
- ☐ $-21 + (-15)$
- ☐ $(-15) + 21$

Select all that apply.

Show your work

calculator · multiselect

13. Which expression is equivalent to $5(x + 6)$?

- A. $x + 30$
- B. $5x + 11$
- C. $5x + 30$
- D. $5x + 6$

Show your work

multiple choice

14. Select **all** the equations that have the solution $x = 6$.

- ☐ $11x = 66$
- ☐ $11x = 6$
- ☐ $x \div 11 = 6/11$
- ☐ $66 = 11x$
- ☐ $x + 11 = 6$
- ☐ $x \div 11 = 66$

Select all that apply.

Show your work

multiselect

15. Which number is located between -4 and -3 on a number line?

- A. $-2 \frac{3}{4}$
- B. $-4 \frac{1}{2}$
- C. $-3 \frac{1}{4}$
- D. -2

Show your work

calculator · multiple choice

16. Use the drop-down menu to complete the statistical question.

How many minutes did [[1]] spend reading last night?

Blank (1) choices: each student in the class | I | my teacher

Show your work

editing task choice

17. Select **all** the expressions equal to $\frac{1}{3}$.

☐ $\frac{2}{3} \div \frac{1}{2}$

☐ $\frac{2}{6}$

☐ $\frac{1}{2} \times \frac{2}{3}$

☐ $\frac{2}{3} + \frac{1}{2}$

☐ $\frac{2}{3} \times \frac{1}{2}$

☐ $\frac{3}{5}$

Select all that apply.

Show your work

calculator · multiselect

18. What is the opposite of $-5\frac{3}{4}$?

A. $-5\frac{3}{4}$

B. $5\frac{3}{4}$

C. $-11\frac{1}{2}$

D. $2\frac{7}{8}$

Show your work

calculator · multiple choice

19. What is the value of $3x^2 - 5$ when $x = 6$?

A. 319

B. 93

C. 31

D. 103

Show your work

multiple choice

20. What is the value of $|-16| - |8|$?

- A. -8
- B. 24
- C. 10
- D. 8

Show your work

calculator · multiple choice

21. Select **all** the expressions with a value of 84.

- ☐ $2^2 \times 3 \times 7$
- ☐ $84 + 1$
- ☐ 84×2
- ☐ 84×1
- ☐ $2 \times 2 \times 3 \times 7$
- ☐ $2^2 \times 3^2 \times 7$

Select all that apply.

Show your work

calculator · multiselect

22. Point M is at (2, -4) and point N is at (6, -4).

What is the distance, in units, between point M and point N?

- A. 4
- B. 8
- C. 6
- D. 2

Show your work

multiple choice

23. Select **all** the equations that have the solution $x = 7.5$.

- ☐ $x + 1.5 = 9$
- ☐ $x - 0.5 = 7$
- ☐ $x - 2 = 10.5$
- ☐ $2x = 15$
- ☐ $x + 1.5 = 7.5$
- ☐ $x \div 2 = 7.5$

Select all that apply.

Show your work

multiselect

24. Select **all** the numbers that are common factors of 24 and 36.

- ☐ 4
- ☐ 3
- ☐ 9
- ☐ 6
- ☐ 2
- ☐ 5

Select all that apply.

Show your work

calculator · multiselect

25. Select **all** the forms that are equivalent to $\frac{3}{20}$.

- ☐ 0.15
- ☐ 15%
- ☐ 1.5
- ☐ $\frac{20}{3}$
- ☐ 150%
- ☐ $\frac{3}{20}$

Select all that apply.

Show your work

calculator · multiselect

26. A point is graphed at $-3\frac{3}{4}$ on a number line.

How many units is the point from 0?

- A. $4\frac{3}{4}$
- B. $3\frac{3}{4}$
- C. $-3\frac{3}{4}$
- D. $1\frac{7}{8}$

Show your work

calculator · multiple choice

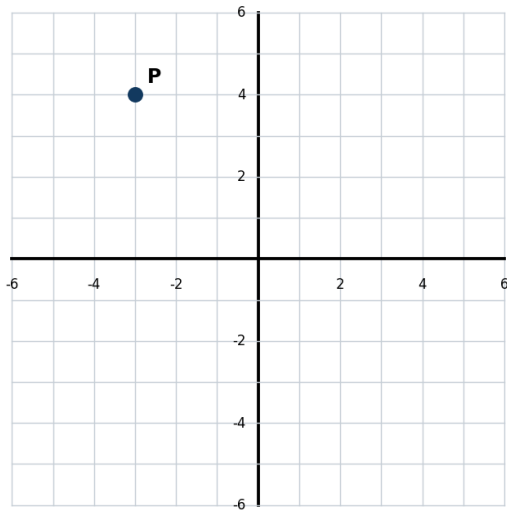
27. Which expression represents a number x decreased by 4?

- A. $x/4$
- B. $4 - x$
- C. $x + 4$
- D. $x - 4$

Show your work

multiple choice

28. What are the coordinates of point P?



- A. (4, -3)
- B. (3, 4)
- C. (-3, -4)
- D. (-3, 4)

Show your work

multiple choice

29. A pitcher holds 8 cups of juice.

Select **all** the expressions that give the number of fluid ounces the pitcher holds. (1 cup = 8 fluid ounces)

- ☒ $8 \div 8$
- ☒ $8 + 8$
- ☒ 8×2
- ☒ 64×1
- ☒ 8×8

Select all that apply.

Show your work

multiselect

30. Select **all** the ratios that belong in a table of ratios equivalent to 5:6.

- ☐ 20:24
- ☐ 6:7
- ☐ 10:6
- ☐ 6:5
- ☐ 5:6
- ☐ 10:12

Select all that apply.

Show your work

multiselect

31. A ticket costs \$7.31 and 6 tickets are bought.

Select **all** the expressions that give the total cost in dollars.

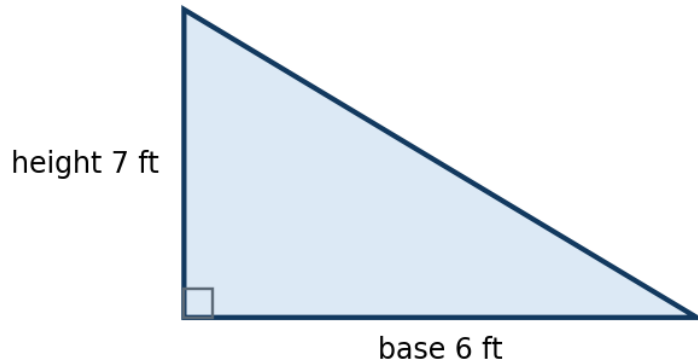
- ☐ 7.31×6
- ☐ $7.31 + 6$
- ☐ 43.86×6
- ☐ $43.86 \div 6$
- ☐ $43.86 \div 1$
- ☐ 6×7.31

Select all that apply.

Show your work

calculator · multiselect

32. What is the area of the right triangle shown, in square feet?



- A. $6 \frac{1}{2}$
- B. 42
- C. 13
- D. 21

Show your work

multiple choice

33. To ride the roller coaster at Tropical Park, a rider must be at least 36 inches tall.

Let x represent the height in inches. Which inequality represents this situation?

- A. $x > 36$
- B. $x \leq 36$
- C. $x < 36$
- D. $x \geq 36$

Show your work

multiple choice

34. What is $2 \frac{2}{3} \div \frac{2}{5}$?

- A. $6 \frac{2}{3}$
- B. $\frac{3}{20}$
- C. $1 \frac{1}{15}$
- D. $3 \frac{1}{15}$

Show your work

calculator · multiple choice

35. Select **all** the expressions equal to $24 + 36$.

- ☐ $13(2 + 3)$
- ☐ $12(2 + 4)$
- ☐ $2(12 + 3)$
- ☐ $12(2 + 3)$
- ☐ 12×5
- ☐ $24 + 36$

Select all that apply.

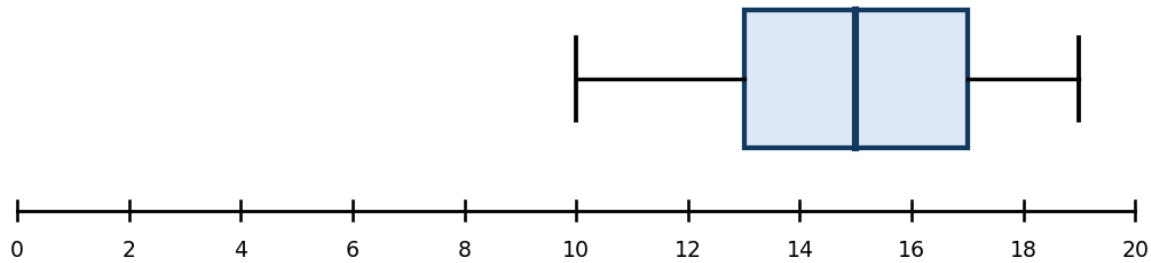
Show your work

calculator · multiselect

36. The box plot shows a set of data.

Select **all** the values that lie between the lower quartile and the upper quartile, inclusive.

Daily high temperature ($^{\circ}\text{F}$)



- ☐ 15
- ☐ 10
- ☐ 17
- ☐ 22
- ☐ 13
- ☐ 19
- ☐ 14

Select all that apply.

Show your work

multiselect

37. Select **all** the expressions that give 60% of 240.

- ☐ 60×240
- ☐ 240×0.6
- ☐ $(60/100) \times 240$
- ☐ $60\% \times 240$
- ☐ $240 \div 60$
- ☐ $240 - 60$

Select all that apply.

Show your work

multiselect

38. Which value of x makes the equation $4x + 10 = 22$ true?

- A. 3
- B. 12
- C. 4
- D. 2

Show your work

multiple choice

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume